SYSTEM RELIABILITY & PROTOCOLS

Safety & Hazard Mitigation

GUIDELINE SPECIFICATION

Physical Fail-Safes

HARDWARE LEVEL

Battery Short Protection
Fuse, emergency cut-off, protected charging

LOW RISK

Water Ingress Defense
Sealed enclosure with strain-relief glands

LOW-MED

Driver Overheating Control
Driver sized to stall current with inline fuse

LOW RISK

Float Leak Prevention
Pre-launch soak testing & initial tethering

LOW RISK

Software Protections

FIRMWARE LEVEL

Signal / Link Loss Timeout
Motors auto-stop after 1s without active command

LOW RISK

Uncommanded Motion Prevention
Motors OFF at boot; strictly accepts valid F/B/L/R/S

LOW RISK

Dual-layer redundancy ensures all residual system risk ratings remain Low to Low-Medium.